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Problem Statement

Question : The conic has vertices $(\pm 6, 0)$ and foci $(\pm 4, 0)$. Find the equation of the conic.

Name	Description	vector form
B₁	vertex 1 of conic	$\begin{pmatrix} 6 \\ 0 \end{pmatrix}$
B₂	vertex 2 of conic	$\begin{pmatrix} -6 \\ 0 \end{pmatrix}$
F₁	focus 1 of conic	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$
F₂	focus 2 of conic	$\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

Table: Vertex and Foci

Solving Conics with Matrices

The conic has two foci , so it cannot be a parabola .

Equation for any conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (4.1)$$

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top \quad (4.2)$$

$$\mathbf{u} = ce^2 \mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F} \quad (4.3)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (4.4)$$

The normal vector of the directrix is along the direction vector of $\mathbf{F}_1 - \mathbf{F}_2$

$$\mathbf{n} = \mathbf{F}_1 - \mathbf{F}_2 \equiv \mathbf{e}_1 \quad (4.5)$$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.6)$$

From (4.2) we can form the matrix $\mathbf{V}$

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - e^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (4.7)$$

$$\mathbf{V} = \begin{pmatrix} 1 - e^2 & 0 \\ 0 & 1 \end{pmatrix} \quad (4.8)$$

As $\mathbf{V}$ is an upper triangular matrix , we get the eigen values as the diagonal entries

$$\lambda_1 = 1 - e^2 \qquad \lambda_2 = 1 \quad (4.9)$$

Clearly $|\mathbf{V}| \neq 0$, $\mathbf{V}^{-1}$ exists.

The center of the conic $\mathbf{c}$ can be found

$$\mathbf{c} = \frac{\mathbf{F}_1 + \mathbf{F}_2}{2} = \mathbf{0} \quad (4.10)$$

The relation between the $\mathbf{c}$, $\mathbf{V}$ and $\mathbf{u}$ is given by

$$\mathbf{V}\mathbf{c} + \mathbf{u} = \mathbf{0} \qquad |\mathbf{V}| \neq 0 \qquad (4.11)$$

$$\mathbf{c} = \mathbf{0} \qquad (4.12)$$

$$\mathbf{u} = \mathbf{0} \qquad (4.13)$$

From (4.3) we get

$$ce^2\mathbf{e}_1 = \mathbf{F}_1 \qquad (4.14)$$

$$\begin{pmatrix} ce^2 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \qquad (4.15)$$

$$ce^2 = 4 \qquad (4.16)$$

$$c = \frac{4}{e^2} \qquad (4.17)$$

$$f_0 = \mathbf{u}^\top \mathbf{V}^{-1} \mathbf{u} - f \qquad (4.18)$$

as $\mathbf{u} = \mathbf{0}$ and from (4.4), we get

$$f_0 = c^2 e^2 - 16 \quad (4.19)$$

The length of the major axis is the distance between the two vertices

$$\|\mathbf{B}_1 - \mathbf{B}_2\| = 12 \quad (4.20)$$

The length of major axes is also given as

$$2\sqrt{\left|\frac{f_0}{\lambda_1}\right|} \quad (4.21)$$

So,

$$2\sqrt{\left|\frac{c^2 e^2 - 16}{1 - e^2}\right|} = 12 \quad (4.22)$$

From (4.17) we get

$$\sqrt{\frac{16}{e^2}} = 6 \quad (4.23)$$

$$\frac{4}{e} = 6 \quad (4.24)$$

$$e = \frac{2}{3} \quad (4.25)$$

As $e < 1$, the conic is an **ellipse**

The value of c and directrix equation are given as

$$c = 9 \quad \mathbf{n}^T \mathbf{x} = \pm 9 \quad (4.26)$$

Using the obtained values of c and e , we get

$$\mathbf{v} = \begin{pmatrix} \frac{5}{9} & 0 \\ 0 & 1 \end{pmatrix} \quad (4.27)$$

$$\mathbf{u} = \mathbf{0} \quad (4.28)$$

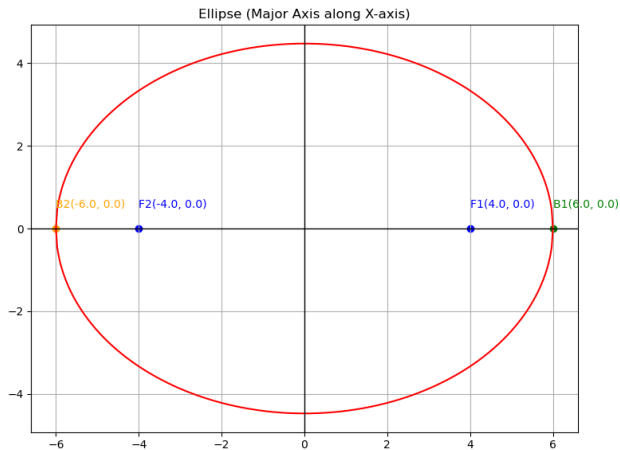
$$(4.29)$$

$$f = -20 \quad (4.30)$$

Substituting these in (4.1) , we get the equation of **ellipse** as

$$\mathbf{x}^\top \begin{pmatrix} \frac{1}{36} & 0 \\ 0 & \frac{1}{20} \end{pmatrix} \mathbf{x} = 1 \quad (4.31)$$

Plots



C Code

```
#include <stdio.h>
#include <math.h>

#define N 400 // Number of points for the ellipse

// Function to generate ellipse coordinates
void ellipse_points(double *x_vals, double *y_upper, double *y_lower,
    double a, double b) {
    double step = (2 * a) / (N - 1); // Step size between x-values

    for (int i = 0; i < N; i++) {
        x_vals[i] = -a + i * step; // X values range from -a to +a
        double value = 1.0 - (x_vals[i] * x_vals[i]) / (a * a);

        if (value < 0)
            value = 0; // Avoid sqrt of small negative rounding errors
    }
}
```

```

        y_upper[i] = b * sqrt(value); // Upper half of ellipse
        y_lower[i] = -y_upper[i]; // Lower half (mirror)
    }
}

// Main function callable from Python
void generate_ellipse(double *x_vals, double *y_upper, double *y_lower) {
    double a = 6.0; // Semi-major axis (x-axis)
    double c = 4.0; // Distance to focus
    double b = sqrt(a * a - c * c); // Semi-minor axis (y-axis)

    ellipse_points(x_vals, y_upper, y_lower, a, b);
}

```

```
import ctypes
import os
import numpy as np

N = 400
# Load the shared object
lib = ctypes.CDLL("./ellipse.so")

# Define argument and return types for the C function
lib.generate_ellipse.argtypes = [
    np.ctypeslib.ndpointer(dtype=np.float64, ndim=1, flags="
        C_CONTIGUOUS"),
    np.ctypeslib.ndpointer(dtype=np.float64, ndim=1, flags="
        C_CONTIGUOUS"),
    np.ctypeslib.ndpointer(dtype=np.float64, ndim=1, flags="
        C_CONTIGUOUS")
]
```

```
lib.generate_ellipse.restype = None
# Create numpy arrays to hold results
x_vals = np.zeros(N, dtype=np.float64)
y_upper = np.zeros(N, dtype=np.float64)
y_lower = np.zeros(N, dtype=np.float64)

# Call the C function
lib.generate_ellipse(x_vals, y_upper, y_lower)

# Print some sample results to verify
print("Sample Ellipse Points:")
for i in range(0, N, 50): # print every 50th point
    print(f'x = {x_vals[i]:.3f}, y_upper = {y_upper[i]:.3f}, y_lower = {
        y_lower[i]:.3f}')

print("\nC function executed successfully.")
```

Python Code for Plotting

```
import numpy as np
import matplotlib.pyplot as plt
import os

# save figure in figs folder
figs_folder = os.path.join("../", "figs")

# Given ellipse parameters
f1 = (4.0, 0.0)
f2 = (-4.0, 0.0)
vert = (6.0, 0.0)
vert2 = (-6.0, 0.0)

# semi-major and semi-minor axes
a = vert[0] # semi-major axis (along x-axis)
c_f = f1[0] # distance to focus from origin
b = np.sqrt(a**2 - c_f**2) # semi-minor axis (along y-axis)
```



```
# x and y values for ellipse
x_vals = np.linspace(-a, a, 400)
y_upper = b * np.sqrt(1 - (x_vals**2 / a**2))
y_lower = -y_upper

# Plotting
fig, ax = plt.subplots(figsize=(8, 8))
ax.plot(x_vals, y_upper, 'r')
ax.plot(x_vals, y_lower, 'r')

# plot foci
ax.scatter([f1[0], f2[0]], [f1[1], f2[1]], color="blue")
ax.text(f1[0], f1[1] + 0.5, f'F1{f1}', fontsize=10, color="blue")
ax.text(f2[0], f2[1] + 0.5, f'F2{f2}', fontsize=10, color="blue")

# plot vertices
ax.scatter([vert[0]], [vert[1]], color="green")
ax.text(vert[0], vert[1] + 0.5, f'B1{vert}', fontsize=10, color="green")
```

```
ax.scatter([vert2[0]], [vert2[1]], color="orange")
ax.text(vert2[0], vert2[1] + 0.5, f' B2{vert2}', fontsize=10, color="
orange")

# center axes
ax.axhline(0, color='black', linewidth=1)
ax.axvline(0, color='black', linewidth=1)

# formatting
ax.set_aspect('equal')
ax.set_title("Ellipse (Major Axis along X-axis)")
ax.grid(True)

# save figure
plt.tight_layout()
fig.savefig(os.path.join(figs_folder, "ellipse.png"))
plt.show()
```